

**SEMINAR SUMMARY:
MARCH 31, 2026 - NEAL BUSHAW
BOOTSTRAP PERCOLATION ON GRIDS.**

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Percolation theory is a branch of mathematics rooted in time based graphical shading changes related to a set of rules and an initial condition. We were presented with an $n \times m$ square tiling of the plane. Our initial condition was a collection of shaded blocks, take Figure ?? for example. The percolating mechanic was then set to being that if a square has k shaded (or infected) neighbors then in the next time step, it will become shaded, and time steps in definite intervals. For example, if $k = 1$, the behaviour is as if each square becomes a “source” in the sense of water or sand, and over time the entire board is filled so long as there is a single shaded block at our initial condition.

We then began looking at the scenario for $k = 2$ (i.e. two neighbors, as edge neighbors, not corner neighbors). If we were to take Figure ??, and step it once in time, then we would get Figure

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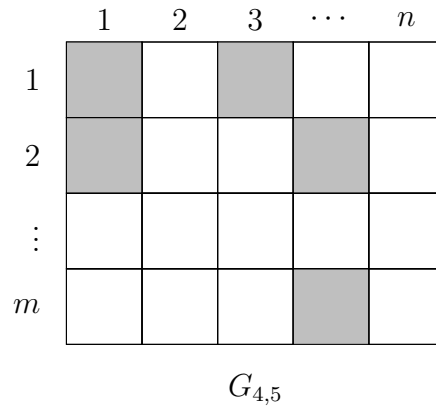


FIGURE 1. A rectangular $G_{4,5}$ tiled with squares, five shaded as bootstrap for percolation.

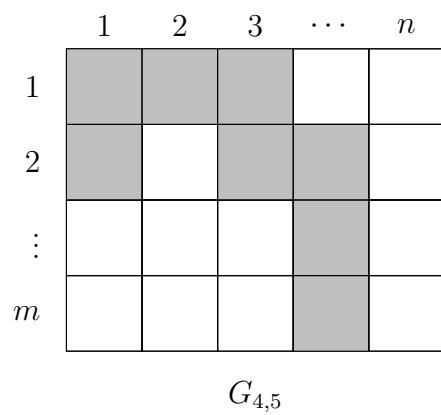


FIGURE 2. A rectangular $G_{4,5}$ tiled with squares, eight shaded after time step 1 with $k = 2$.